Matrix

矩陣計算

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(1) Matrix multiplication

(2) $AX = B$

(3) $WX + B = S$

(1) Vector: $V = \begin{bmatrix} 3 & 0 & -1 \end{bmatrix}$

(2) Matrix: $M_1 = \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

$$M_2 = \begin{bmatrix} -2 & 3 \\ 4 & 0 \\ 1 & -5 \end{bmatrix}$$

$$\underset{2\text{x}2}{\mathrm{A}} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad \underset{2\text{x}3}{B} = \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

row

$$\underset{2\text{x}2}{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

column

row

$$\underset{2\text{x}3}{B} = \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

column

row

$$\underset{2\text{x}2}{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

column

row

$$\underset{2\text{x}3}{B} = \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

column

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

2x2 2x3

$$= \begin{bmatrix} 5 & -2 & 3 \\ & & \end{bmatrix}$$

$1 \cdot 3 + 2 \cdot 1 = 5$

$1 \cdot 0 + 2 \cdot (-1) = -2$

$1 \cdot (-1) + 2 \cdot 2 = 3$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

2x2 2x3

$$= \begin{bmatrix} 5 & -2 & 3 \\ & & \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -2 & 3 \\ 13 & -4 & 5 \end{bmatrix}$$

$3 \cdot 3 + 4 \cdot 1 = 13$

$3 \cdot 0 + 4 \cdot (-1) = -4$

$3 \cdot (-1) + 4 \cdot 2 = 5$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

2x2 2x3

2x3

$$= \begin{bmatrix} 5 & -2 & 3 \\ 13 & -4 & 5 \end{bmatrix}$$

2x3

$$BA = \begin{bmatrix} 3 & 0 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

2x3 2x2

$3 \neq 2$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 13 & -4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 13 & -4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ -2 & -2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 13 & -4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ -2 & -2 \end{bmatrix}$$

Transpose:

(exchange row and column)

$$M_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \qquad M_1^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Transpose:

(exchange row and column)

$$M_2 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$M_2^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Matrix Inversion:

$$\mathrm{C} = \begin{bmatrix} 3 & 6 \\ -2 & -2 \end{bmatrix}$$

Matrix Inversion:

$$C = \begin{bmatrix} 3 & 6 \\ -2 & -2 \end{bmatrix}, \qquad C^{-1} = \begin{bmatrix} -1/3 & -1 \\ 1/3 & 1/2 \end{bmatrix}$$

Matrix Inversion:

$$C = \begin{bmatrix} 3 & 6 \\ -2 & -2 \end{bmatrix}, \qquad C^{-1} = \begin{bmatrix} -1/3 & -1 \\ 1/3 & 1/2 \end{bmatrix}$$

$$CC^{-1} = C^{-1}C = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

C^{-1}: inverse of C, $\qquad I_n$: identy matrix

jupyter

In [1]:
```
import numpy as np

A1 = np.matrix('1 2; 3 4')
print(A1)
```

```
[[1 2]
 [3 4]]
```

In [2]:
```
B1 = np.matrix('3 0 -1; 1 -1 2')
print(B1)
```

```
[[ 3  0 -1]
 [ 1 -1  2]]
```

In [3]:
```
C1 = A1*B1
print(C1)
```

```
[[ 5 -2  3]
 [13 -4  5]]
```

jupyter

In [4]:
```
A2 = np.matrix('1 2; 3 4')
print(A2)
B2 = np.matrix('3 0; 1 -1')
print(B2)
C2 = A2*B2
print(C2)
```

```
[[1 2]
 [3 4]]
[[ 3  0]
 [ 1 -1]]
[[ 5 -2]
 [13 -4]]
```

In [5]:
```
D2 = B2*A2
print(D2)
```

```
[[ 3  6]
 [-2 -2]]
```

jupyter

In [5]:
```
D2 = B2*A2
print(D2)
```

```
[[ 3  6]
 [-2 -2]]
```

In [6]:
```
Dinv = np.linalg.inv(D2)
print(Dinv)
```

```
[[-0.33333333 -1.        ]
 [ 0.33333333  0.5       ]]
```

(1) Matrix multiplication

(2) $AX = B$

(3) $WX + B = S$

AX=B: a system of linear equations

$$2x_1 + x_2 - x_3 = 0$$

$$x_1 - 2x_2 + x_3 = 7$$

$$-x_1 + 2x_2 + 3x_3 = 5$$

$$x_1 = 2, x_2 = -1, x_3 = 3$$

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & -2 & 1 \\ -1 & 2 & 3 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 7 \\ 5 \end{bmatrix}$$

$$AX = B$$

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 0 \\ x_1 - 2x_2 + x_3 &= 7 \\ -x_1 + 2x_2 + 3x_3 &= 5 \end{aligned}$$

$$\underset{3\text{x}3}{A} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & -2 & 1 \\ -2 & 2 & 3 \end{bmatrix}, \underset{3\text{x}1}{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \underset{3\text{x}1}{B} = \begin{bmatrix} 0 \\ 7 \\ 5 \end{bmatrix}$$

$$\underset{3\text{x}3\ 3\text{x}1}{AX} = \underset{3\text{x}1}{B}$$

3x1

$$2x_1 + x_2 - x_3 = 0$$
$$x_1 - 2x_2 + x_3 = 7$$
$$-x_1 + 2x_2 + 3x_3 = 5$$

$$\mathrm{A}\,\mathrm{X} = B$$

3x3 3x1 3x1

$$\underset{3\text{x}3}{\mathrm{A}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}1}{B}$$

$$\mathrm{A}^{-1}\,\mathrm{A}\,\mathrm{X} = \mathrm{A}^{-1}B$$

$$\underset{3\text{x}3}{\mathrm{A}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}1}{B}$$

$$\underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}3}{\mathrm{A}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}1}{B}$$

$$\underset{3\text{x}3}{\mathrm{A}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}1}{B}$$

$$\underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}3}{\mathrm{A}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}1}{B}$$

$$\mathrm{A}^{-1}\mathrm{A} = I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\underset{3\text{x}3}{\mathrm{A}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}1}{B}$$

$$\boxed{\underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}3}{\mathrm{A}}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}1}{B}$$

$$\boxed{\underset{3\text{x}3}{\mathrm{I}}}\ \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}1}{B}$$

$$\underset{3\text{x}1}{\mathrm{X}} = \underset{3\text{x}3}{\mathrm{A}^{-1}}\ \underset{3\text{x}1}{B}$$

$$\underset{3\text{x}3}{A}\ \underset{3\text{x}1}{X} = \underset{3\text{x}1}{B} \quad \Rightarrow \quad \underset{3\text{x}1}{X} = \underset{3\text{x}3}{A^{-1}}\ \underset{3\text{x}1}{B}$$

$$X = A^{-1}B = \frac{1}{20}\begin{bmatrix} 8 & 5 & 1 \\ 4 & -5 & 3 \\ 0 & 5 & 5 \end{bmatrix}\begin{bmatrix} 0 \\ 7 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

jupyter

In [7]:
```
X3 = np.matrix('2; -1; 3')
A3 = np.matrix('2 1 -1; 1 -2 1; -1 2 3')
B3 = A3*X3
print('A3=',A3)
print('X3=',X3)
print('B3=',B3)
A3inv = np.linalg.inv(A3)
print('A3inv=',A3inv)
X4 = A3inv*B3
print('X4=',X4)
```

jupyter

```
A3= [[ 2  1 -1]
 [ 1 -2  1]
 [-1  2  3]]
X3= [[ 2]
 [-1]
 [ 3]]
B3= [[0]
 [7]
 [5]]
A3inv= [[ 0.4   0.25  0.05]
 [ 0.2  -0.25  0.15]
 [ 0.    0.25  0.25]]
X4= [[ 2.]
 [-1.]
 [ 3.]]
```

AX=B: a system of linear equations

$$2x_1 + x_2 - x_3 = 0$$

$$x_1 - 2x_2 + x_3 = 7$$

$$-x_1 + 2x_2 + 3x_3 = 5$$

$$x_1 = 2, x_2 = -1, x_3 = 3$$

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots \qquad\qquad\qquad\qquad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

n: unknowns

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

$$\vdots \qquad\qquad \vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

m: equations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \qquad\qquad\qquad\qquad & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

m: equations

n: unknowns

m rows

n columns

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \vdots \qquad\qquad\qquad & \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

m: equations

n: unknowns

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$\underset{m\text{x}n}{\mathrm{A}} \qquad \underset{n\text{x}1}{\mathrm{X}} = \underset{m\text{x}1}{\mathrm{B}}$$

$$\underset{m\times n}{A}\ \underset{n\times 1}{X} = \underset{m\times 1}{B} \quad \Rightarrow \quad X = ?$$

(1) m (equations) = n (unknowns), unique solution

(2) m (equations) < n (unknowns), many solutions

(3) m (equations) > n (unknowns), least-square solution

$$A\,X = B \quad \Rightarrow \quad X = ?$$

mxn nx1 mx1

600x2 2x1 600x1

(1) m (equations) = n (unknowns), unique solution

(2) m (equations) < n (unknowns), many solutions

(3) m (equations) > n (unknowns), least-square solution

$$A\,X = B$$

600x2, 2x1 600x1

$$A\,X = \qquad\qquad B$$

600x2, 2x1 600x1

$$A^T\,A\,X = \qquad\qquad A^T\,B$$

2x600, 600x2, 2x1 2x600, 600x1

$$A\,X = B$$

600x2, 2x1 600x1

$$A^T\,A\,X = A^T\,B$$

2x600, 600x2, 2x1 2x600, 600x1

$$(A^T A)^{-1}\,A^T\,A\,X = (A^T A)^{-1}\,A^T\,B$$

2x2 2x2, 2x1 2x2, 2x600, 600x1

$$A\,X = B$$

600x2, 2x1 600x1

$$A^T\,A\,X = A^T\,B$$

2x600, 600x2, 2x1 2x600, 600x1

$$(A^T A)^{-1}\,A^T\,A\,X = (A^T A)^{-1}\,A^T\,B$$

2x2 2x2, 2x1 2x2, 2x600, 600x1

$$I\,X = (A^T A)^{-1}\,A^T\,B$$

2x2, 2x1 2x2, 2x600, 600x1

$$A\,X = B$$

600x2, 2x1 600x1

$$A^T\,A\,X = A^T\,B$$

2x600, 600x2, 2x1 2x600, 600x1

$$(A^T A)^{-1}\,A^T\,A\,X = (A^T A)^{-1}\,A^T\,B$$

2x2 2x2, 2x1 2x2, 2x600, 600x1

$$I\,X = (A^T A)^{-1}\,A^T\,B$$

2x2, 2x1 2x2, 2x600, 600x1

$$X = A^+ \quad B$$

2x1 2x600 600x1

(3) m (equations) > n (unknowns)

$$\underset{n\times m}{A^{+}} = \underbrace{(\underset{n\times m}{A^{T}}\,\underset{m\times n}{A})^{-1}}_{n\times n}\,\underset{n\times m}{A^{T}}$$

Pseudo-inverse (Moore–Penrose inverse)

A^{T} is the transpose of A

$$\underset{n\times 1}{\hat{X}} = \underset{n\times m}{A^{+}}\,\underset{m\times 1}{B}$$

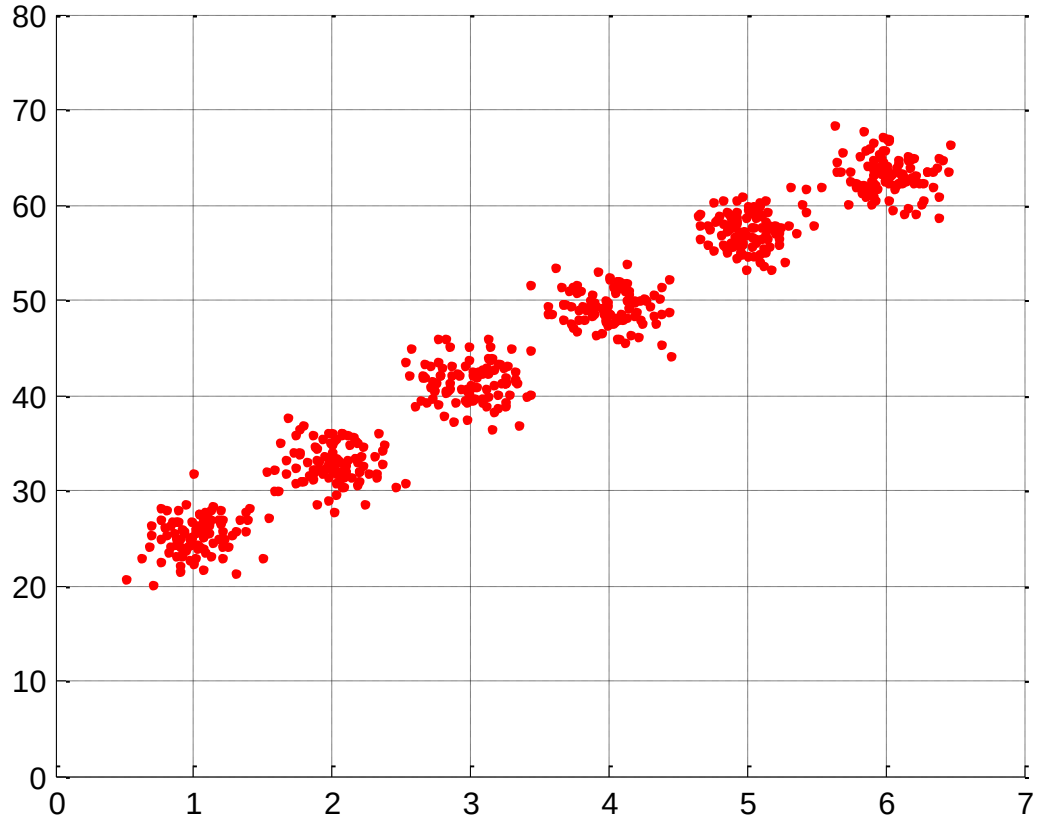
Estimated X

An example, curve fitting of a 600 points data to a linear equation

$y = mx + b$

Find the best Estimate of

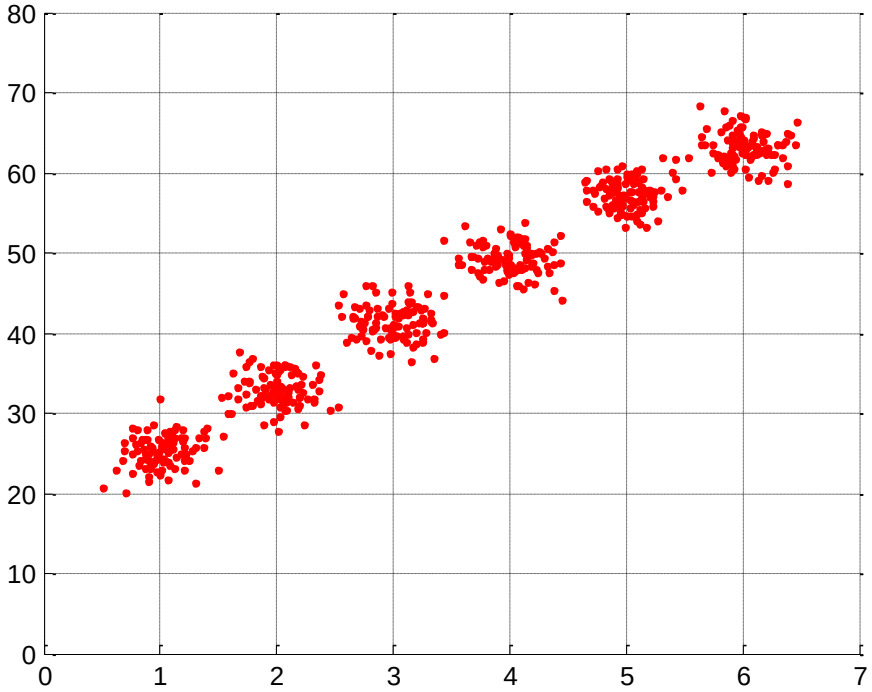
m and b



(3) N (equations) > n (unknowns)

$$An\ example, a\ dataset\ of\ N = 600\ points:$$

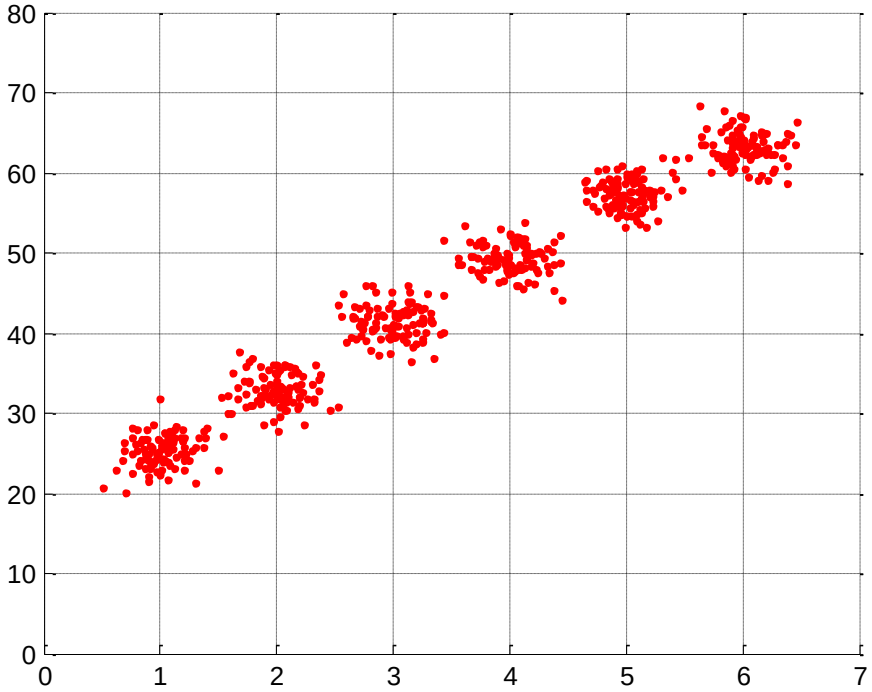
$$(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N), N = 600$$



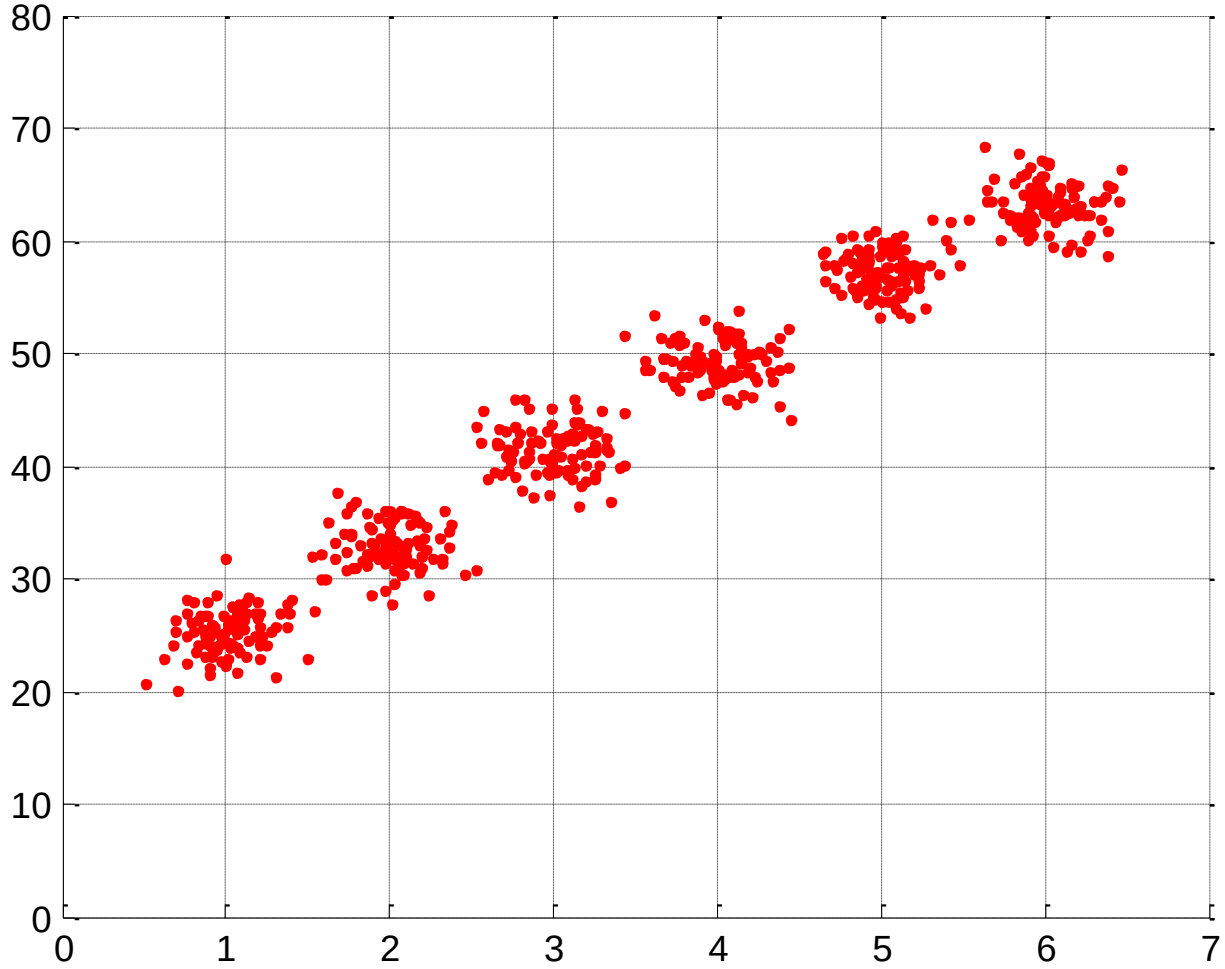
(3) N (equations) > n (unknowns)

$$To\ fit\ the\ linear\ equation: y = mx + b$$

$$Find\ the\ best\ estimate:\ \ \hat{m}, \hat{b}$$



(3) N (equations) > n (unknowns)



$$mx_1 + b = y_1$$

$$mx_2 + b = y_2$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$mx_N + b = y_N$$

(3) N (equations) > n (unknowns)

$$mx_1 + b = y_1$$

$$mx_2 + b = y_2$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$mx_N + b = y_N$$

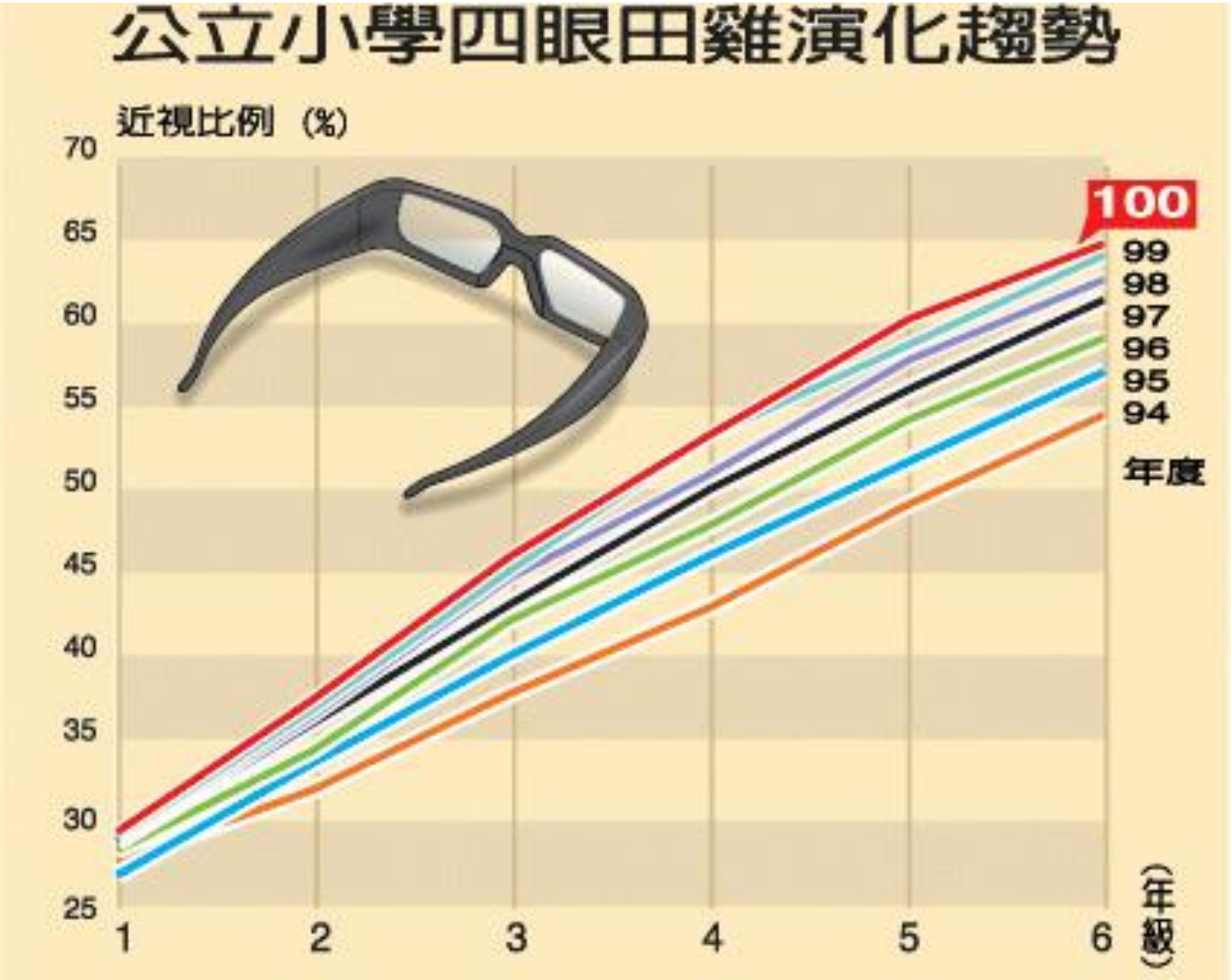
$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ . & . \\ . & . \\ x_N & 1 \end{bmatrix}}_{\substack{A \\ 600\text{x}2}} \underbrace{\begin{bmatrix} m \\ b \end{bmatrix}}_{\substack{X \\ 2\text{x}1}} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}}_{\substack{B \\ 600\text{x}1}}$$

(3) N (equations) > n (unknowns)

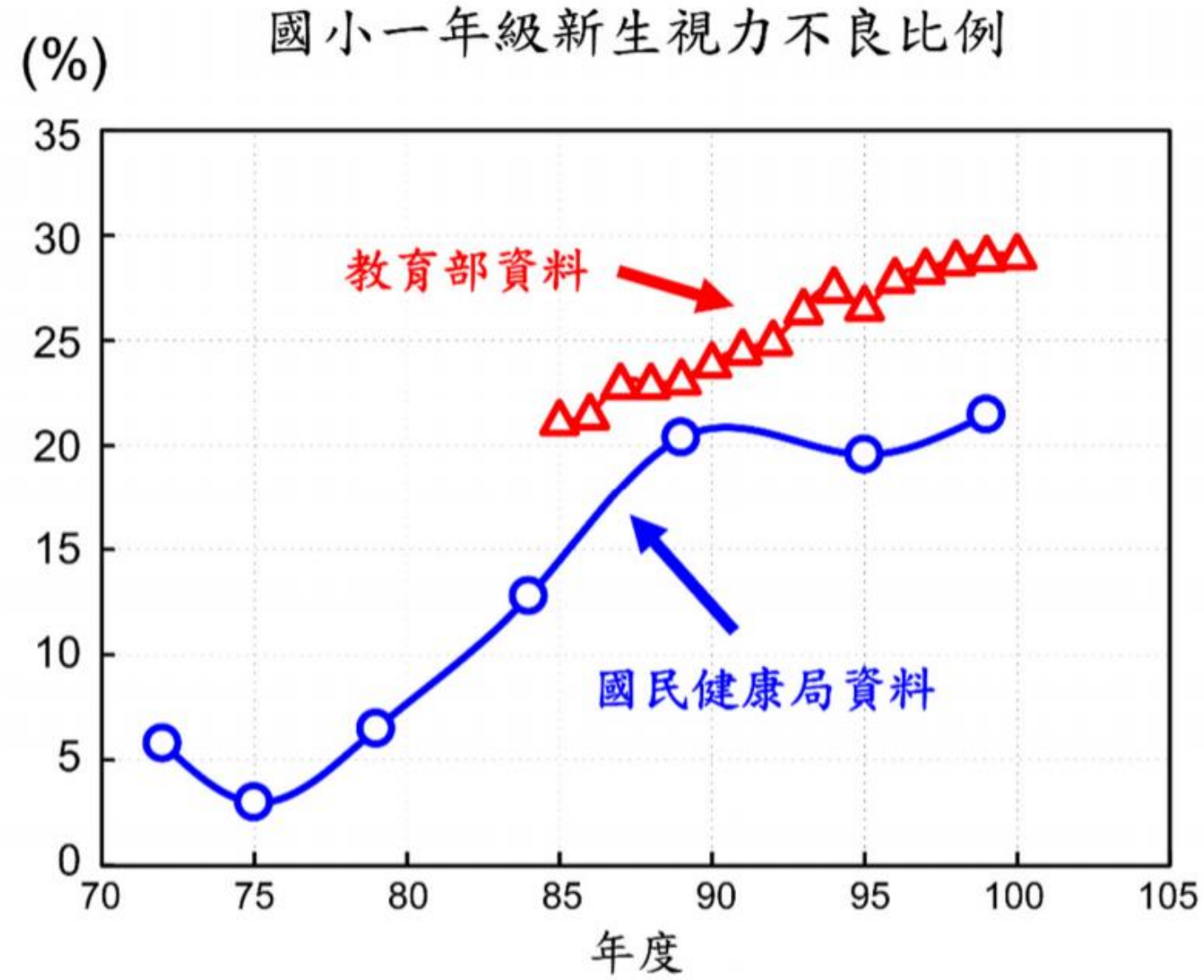
$$\underbrace{\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ . & . \\ . & . \\ x_N & 1 \end{bmatrix}}_{\substack{A \\ 600\text{x}2}} \underbrace{\begin{bmatrix} m \\ b \end{bmatrix}}_{\substack{X \\ 2\text{x}1}} = \underbrace{\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}}_{\substack{B \\ 600\text{x}1}}$$

$$\underset{n\text{x}1}{\hat{X}} = \underset{n\text{x}m}{A^{+}} \underset{m\text{x}1}{B}$$

$$\underset{n\text{x}m}{A^{+}} = \underbrace{(\underset{n\text{x}m}{A^{T}} \underset{m\text{x}n}{A})^{-1}}_{n\text{x}n} \underset{n\text{x}m}{A^{T}}$$



https://sites.google.com/view/yhkao/報紙投書/ipad-取代教科書-搶救四眼小低頭族



https://sites.google.com/view/yhkao/報紙投書/兒童節最好的禮物-視力急惡化-搶救小四眼田雞

中華民國統計資訊網
National Statistics

國小學生視力不良率

- 變數與數值

編輯與計算 | 另存新檔 | 繪圖與地圖
捷徑 | XLS ODS CSV PX M+ | 繪圖教學

學生視力不良率-年 依 期間, 指標, 年級別 與 性別

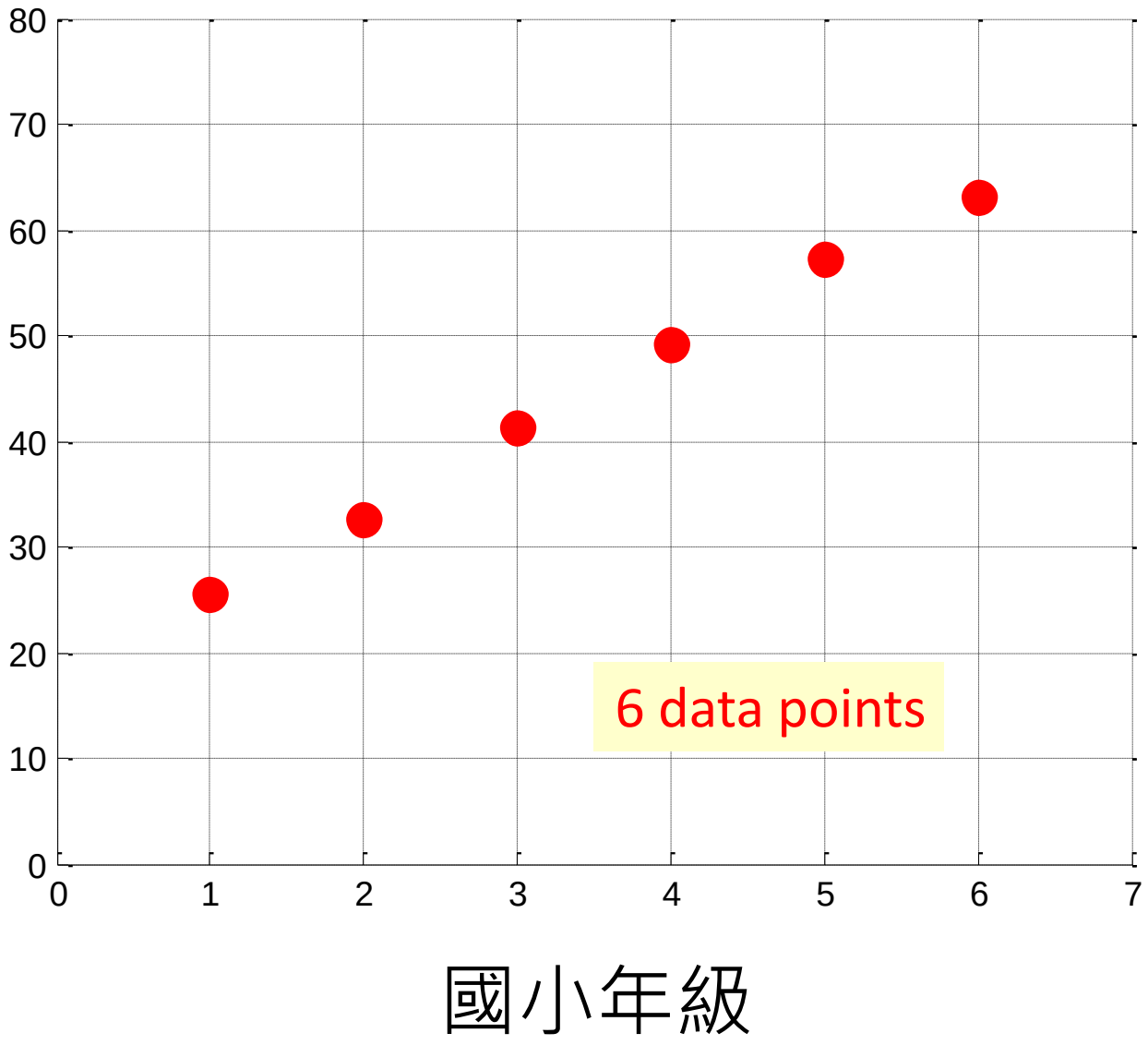
	國小一年級			國小二年級			國小三年級			國小四年級			國小五年級			國小六年級		
	計	男	女	計	男	女	計	男	女	計	男	女	計	男	女	計	男	女
2017	25.64	25.11	26.22	32.71	32.39	33.07	41.33	41.23	41.44	49.13	48.31	50.04	57.21	55.90	58.64	63.21	61.20	65.43

註解：
more
國情統計通報
1.兩眼視力均達0.9以上者為視力正常；一眼在0.9以下者為視力不良。
2.高中(職)自2014學年度起加入實用技能學程學生。

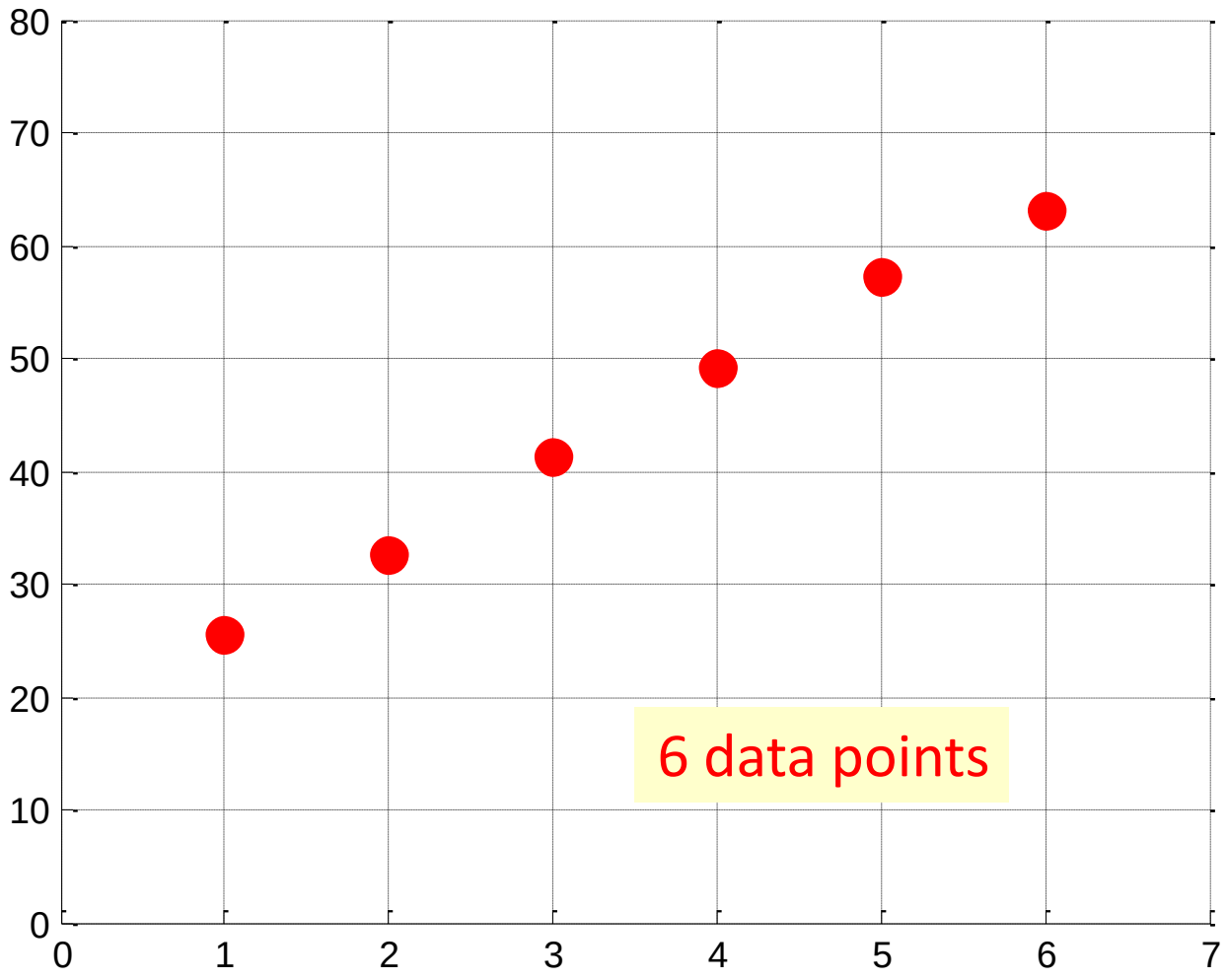
最近更新日：2018-04-30 18:00

http://statdb.dgbas.gov.tw/pxweb/Dialog/Saveshow.asp

國小學生視力不良率



年級	視力不良率(%)
1	25.64
2	32.71
3	41.33
4	49.13
5	57.21
6	63.21



國小年級

Curve fitting to a linear equation:

$$y = mx + b$$

jupyter Python_Pseudo_Inverse_Near_Sightedness

File Edit View Insert Cell Kernel Widgets Help

Run Code

國小兒童近視比例 - 第1個方法

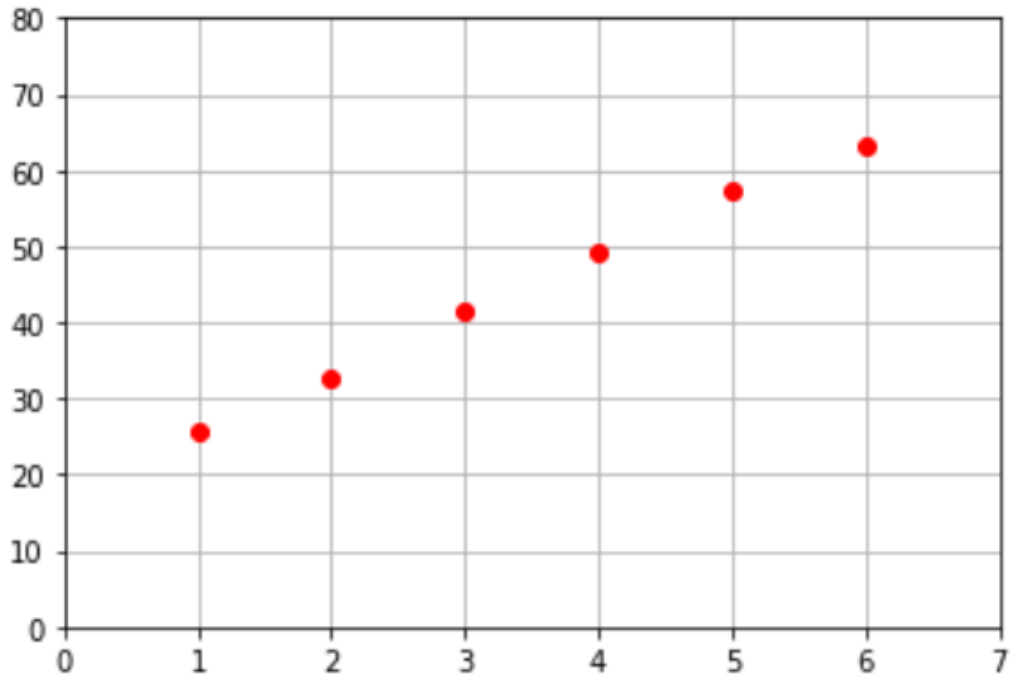
Pseudo inverse

In [1]:
```
import math
import numpy as np
import matplotlib.pyplot as plt
```

輸入資料

In [2]:
```
x = np.array([1.0,2.0,3.0,4.0,5.0,6.0])
y = np.array([25.64,32.71,41.33,49.13,57.21,63.21])
print(x.dtype, x.size, y.dtype, y.size)
plt.plot(x,y,'ro')
plt.xlim([0,7])
plt.ylim([0,80])
plt.grid()
```

```
float64 6 float64 6
```



設定 A_matrix and B_matrix

In [5]:
```
only_one = np.ones(x.size)
A_array = (x,only_one)
A_tmp = np.asmatrix(A_array)
A_matrix = np.matrix.transpose(A_tmp)

B_tmp = np.asmatrix(y)
B_matrix = np.matrix.transpose(B_tmp)

print(A_matrix,'\n')
print(A_matrix.dtype,A_matrix.size,'\n')
print(B_matrix,'\n')
print(B_matrix.dtype,B_matrix.size,'\n')
```

```
[[1. 1.]
 [2. 1.]
 [3. 1.]
 [4. 1.]
 [5. 1.]
 [6. 1.]]

float64 12

[[25.64]
 [32.71]
 [41.33]
 [49.13]
 [57.21]
 [63.21]]

float64 6
```

做 Pseudo inverse, 計算 X_matrix

In [6]:
```
A_transpose = np.matrix.transpose(A_matrix)
A_square = A_transpose*A_matrix
A_square_inverse = np.linalg.inv(A_square)

X_matrix = A_square_inverse * A_transpose * B_matrix
print(A_square,'\n')
print(A_square_inverse,'\n')
print(X_matrix,'\n')
```

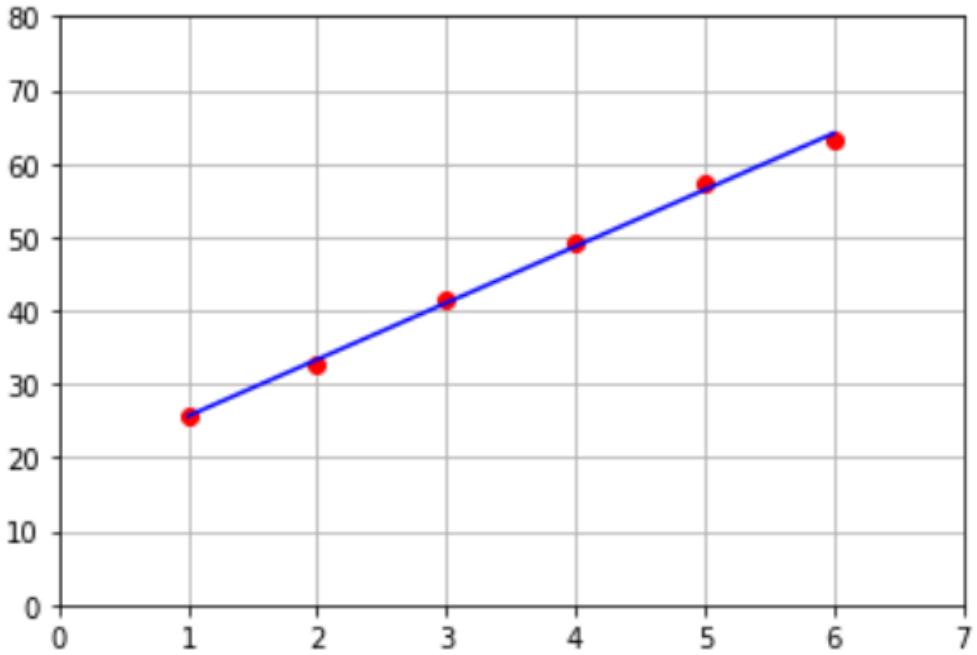
```
[[91. 21.]
 [21.  6.]]

[[ 0.05714286 -0.2        ]
 [-0.2         0.86666667]]

[[ 7.69      ]
 [17.95666667]]
```

列印結果

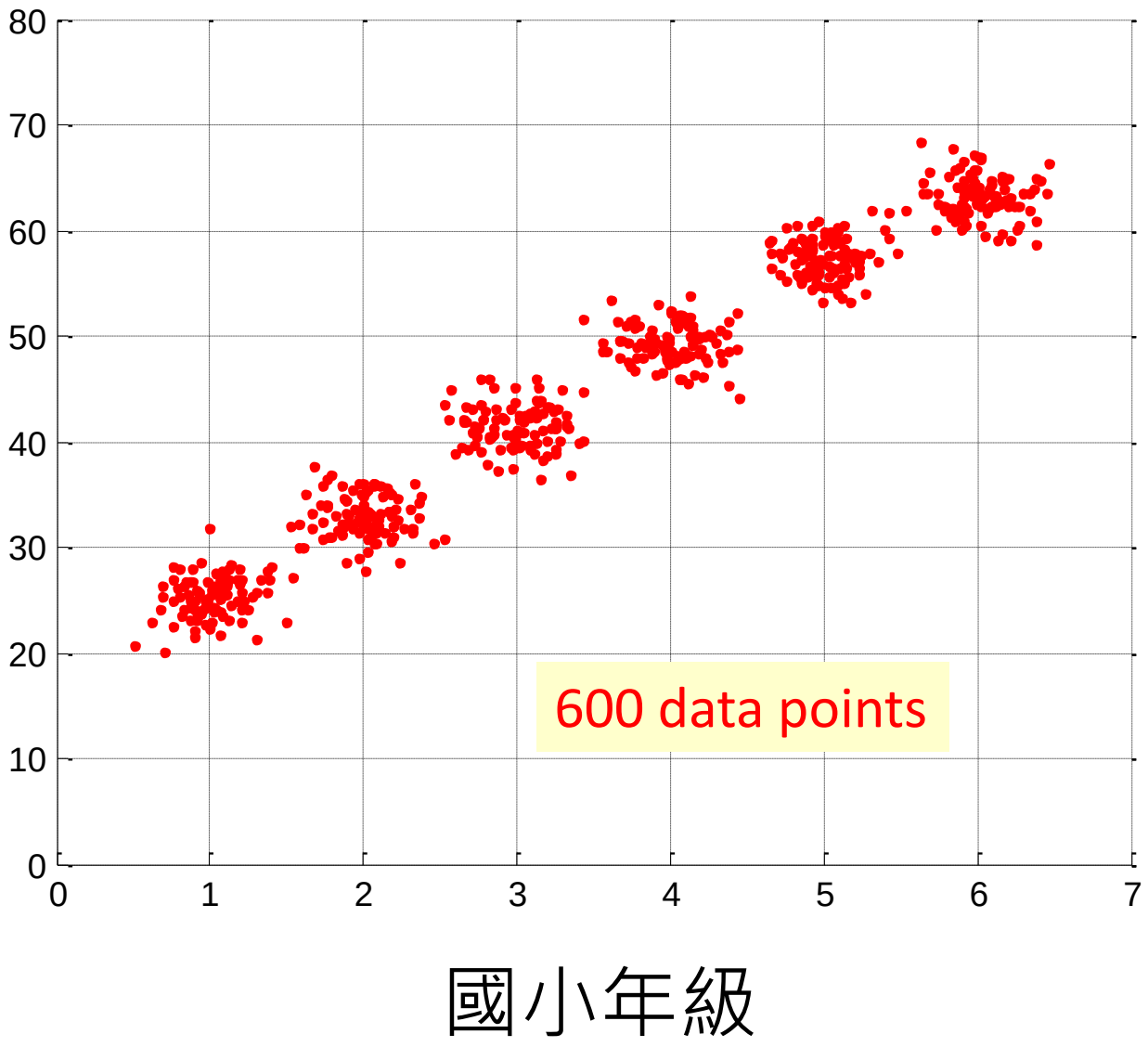
Plot the fitted line

In [7]:
```
x_fit = np.zeros(2)
y_fit = np.zeros(2)
x_fit[0] = 1
x_fit[1] = 6
m_slope = X_matrix[0]
b_bias = X_matrix[1]
y_fit[0] = m_slope*1 + b_bias
y_fit[1] = m_slope*6 + b_bias
```

In [8]:
```
plt.xlim([0,7])
plt.ylim([0,80])
plt.plot(x,y,'ro')
plt.plot(x_fit,y_fit,'b-')
plt.grid()
plt.show()
```



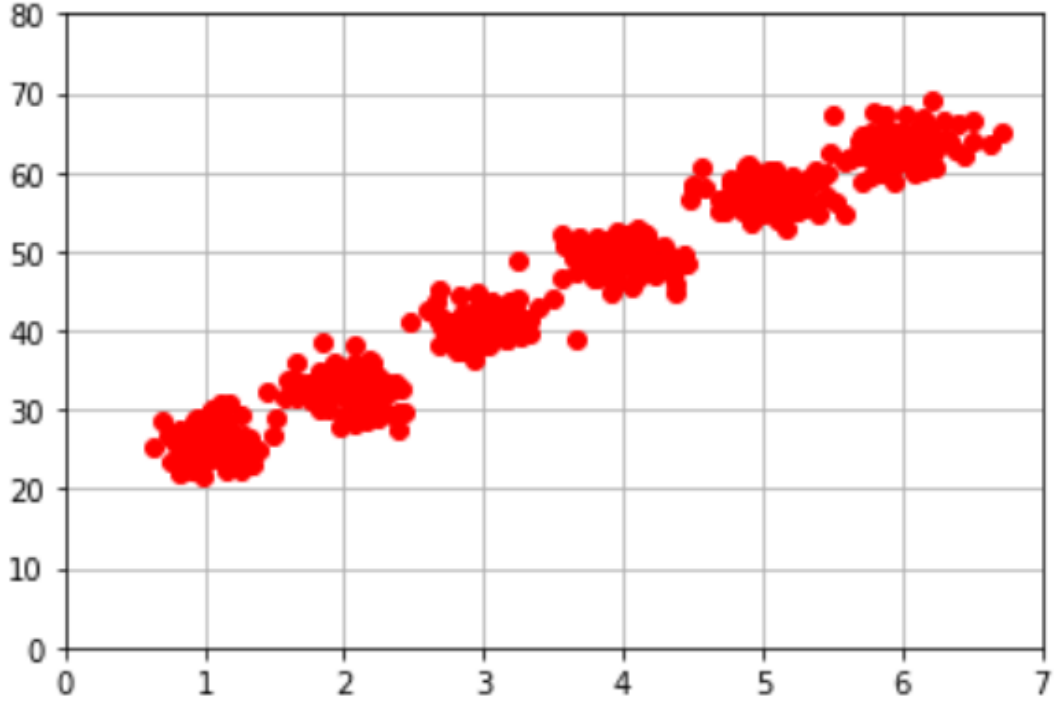
國小學生視力不良率



年級	視力不良率(%)
1	25.64
2	32.71
3	41.33
4	49.13
5	57.21
6	63.21

製造雜訊

In [3]:
```
mu, sigma = 0, 1 # mean and standard deviation
x_noise = 0.2*np.random.normal(mu, sigma, 600)
y_noise = 2.0*np.random.normal(mu, sigma, 600)
```



將雜訊加到原始數據

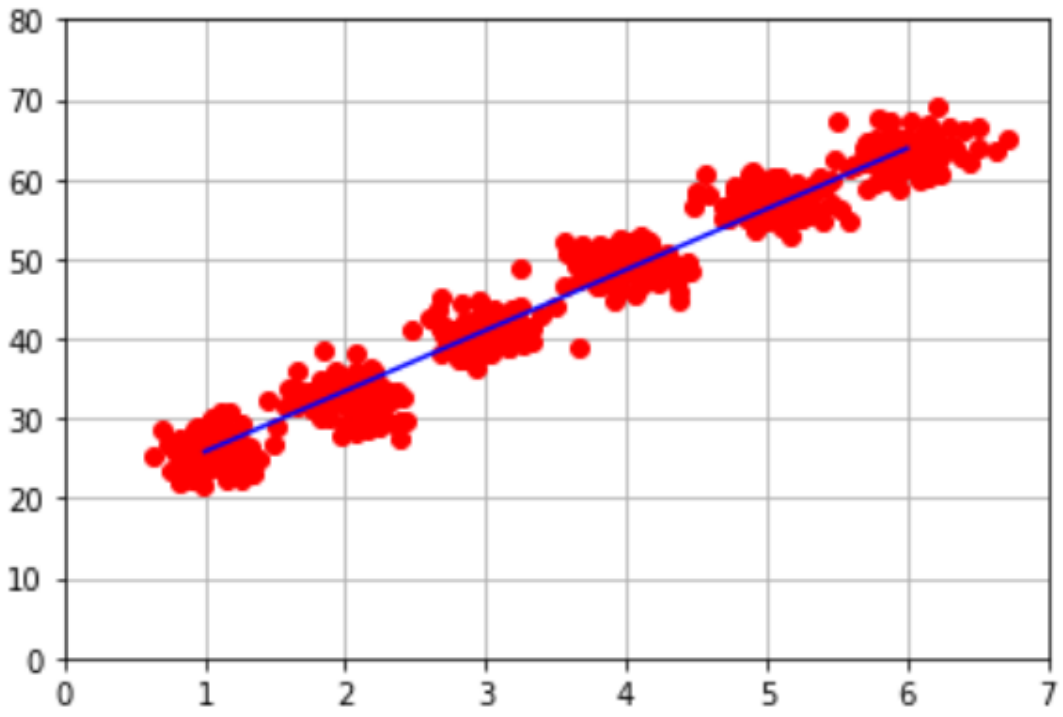
In [4]:
```
x_new = np.zeros(600)
y_new = np.zeros(600)
for i in range(0,600):
    r1 = int(i/100)
    x_new[i] = x[r1] + x_noise[i]
    y_new[i] = y[r1] + y_noise[i]

#x = x_new
#y = y_new
x = x_new
y = y_new
print(x.dtype, x.size, y.dtype, y.size)

plt.plot(x,y,'ro')
plt.xlim([0,7])
plt.ylim([0,80])
plt.grid()
```

```
float64 600 float64 600
```

設定 A_matrix and B_matrix

In [5]:
```
only_one = np.ones(x.size)
A_array = (x,only_one)
A_tmp = np.asmatrix(A_array)
A_matrix = np.matrix.transpose(A_tmp)

B_tmp = np.asmatrix(y)
B_matrix = np.matrix.transpose(B_tmp)

#print(A_matrix,'\n')
print(A_matrix.dtype,A_matrix.size,'\n')
#print(B_matrix,'\n')
print(B_matrix.dtype,B_matrix.size,'\n')
```

```
float64 1200

float64 600
```



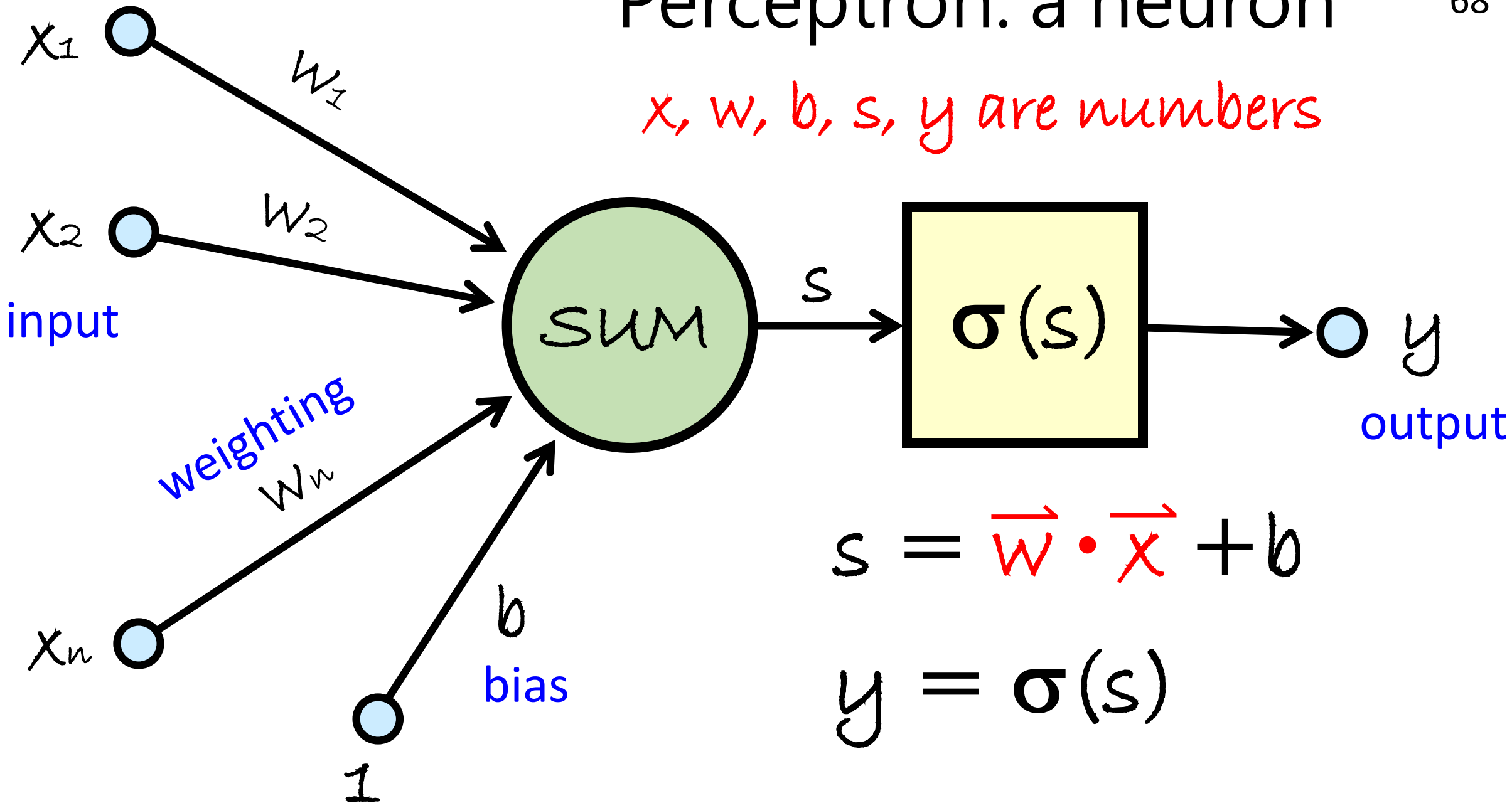
(1) Matrix multiplication

(2) $AX = 0$

(3) $WX + B = S$

Perceptron: a neuron

x, w, b, s, y are numbers



$$s = \vec{w} \cdot \vec{x} + b$$

$$y = \sigma(s)$$

$$s = \vec{w} \cdot \vec{x} + b$$

$$\vec{w} = (w_1, w_2, \ldots, w_n)$$

$$\vec{x} = (x_1, x_2, \ldots, x_n)$$

$$\vec{w} \cdot \vec{x} = w_1x_1 + w_2x_2 + \ldots + w_nx_n$$

$$\underset{1\text{x}n}{\mathrm{W}} = [\, w_1, \quad w_2, \quad \dots \;, \quad w_n], \qquad \underset{n\text{x}1}{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\underset{1\text{x}n \;\; n\text{x}1}{\mathrm{W\,X}} = w_1\, x_1 + w_2 x_2 + \;\dots\; + w_n x_n$$

1x1

$$s = \vec{w} \cdot \vec{x} + b$$

$$= \underset{1\text{x}n \quad n\text{x}1}{W\,X} + b$$

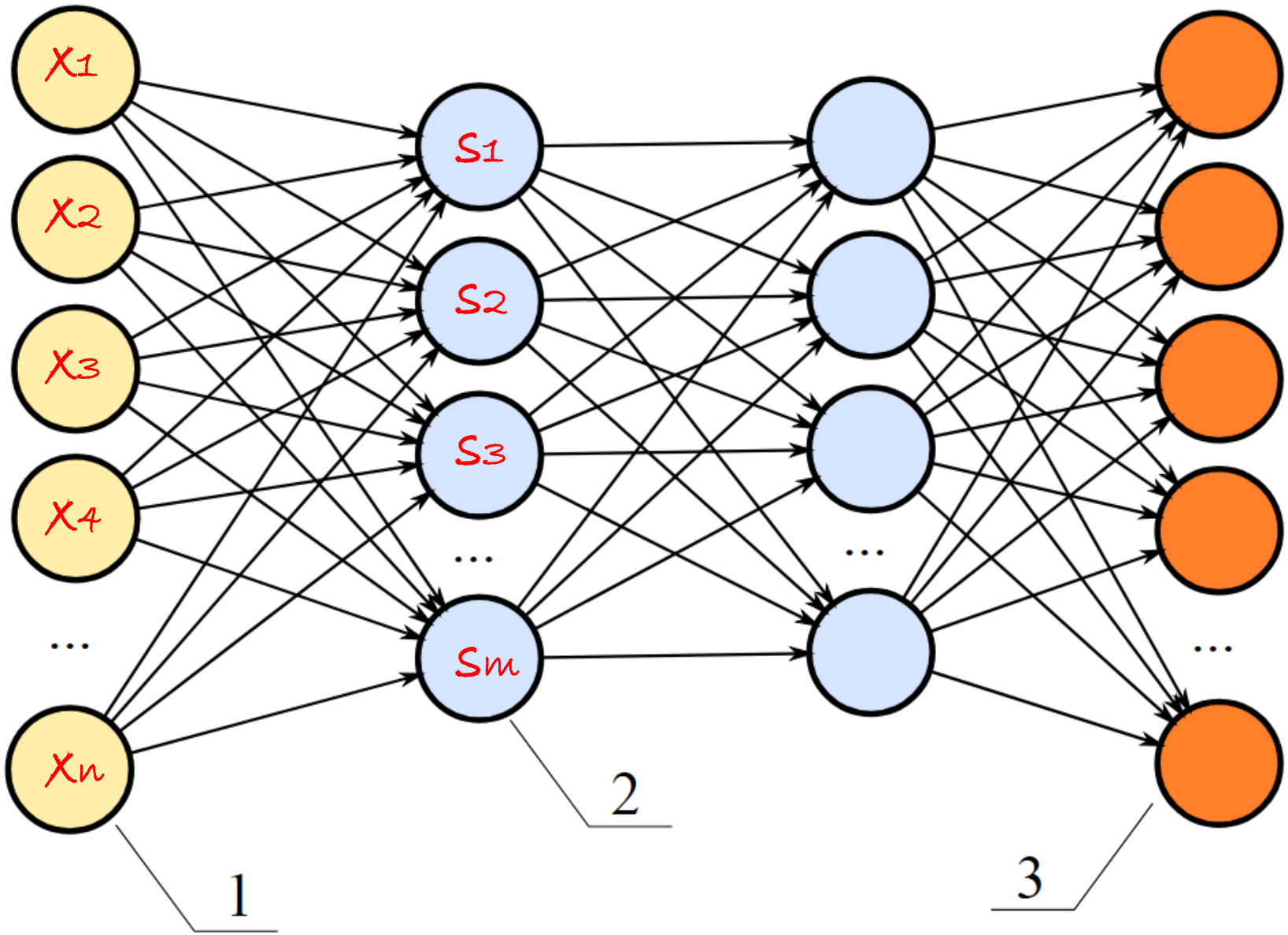
$$= w_1 x_1 + w_2 x_2 + \ \ldots \ + w_n x_n + b$$

$$s = \vec{w} \cdot \vec{x} + b$$

$$= \underset{1\text{x}n \quad n\text{x}1}{\mathrm{W}\,\mathrm{X}} + \mathrm{b}$$

$$= w_1 x_1 + w_2 x_2 + \ \ldots \ + w_n x_n + b$$

$$w_1 x_1 + w_2 x_2 + \ \ldots \ + w_n x_n + b = s$$



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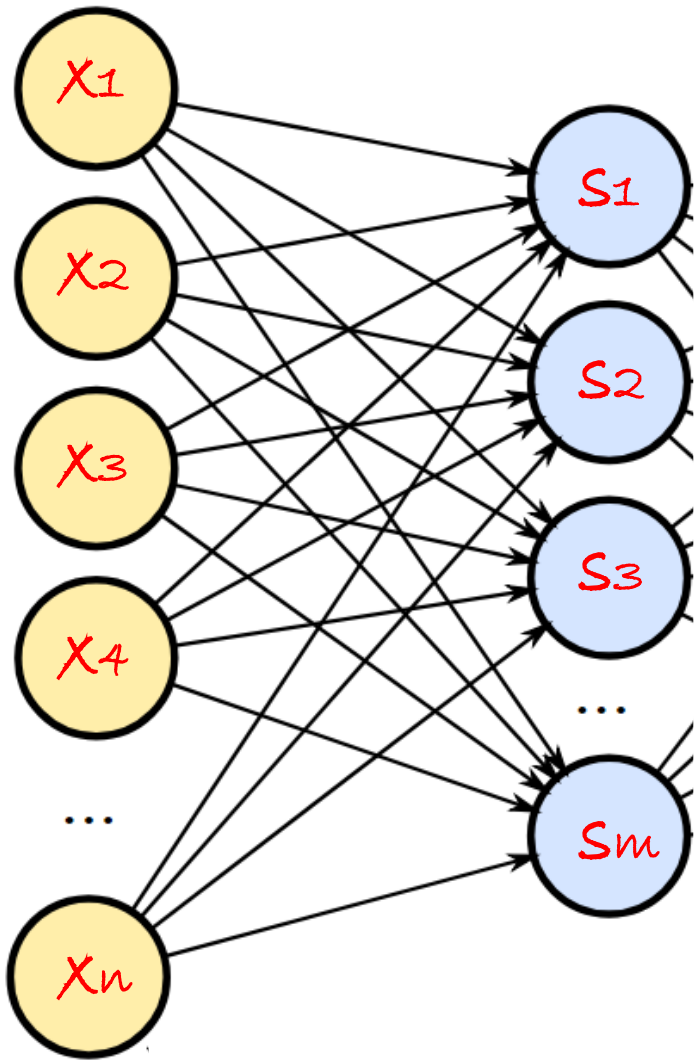

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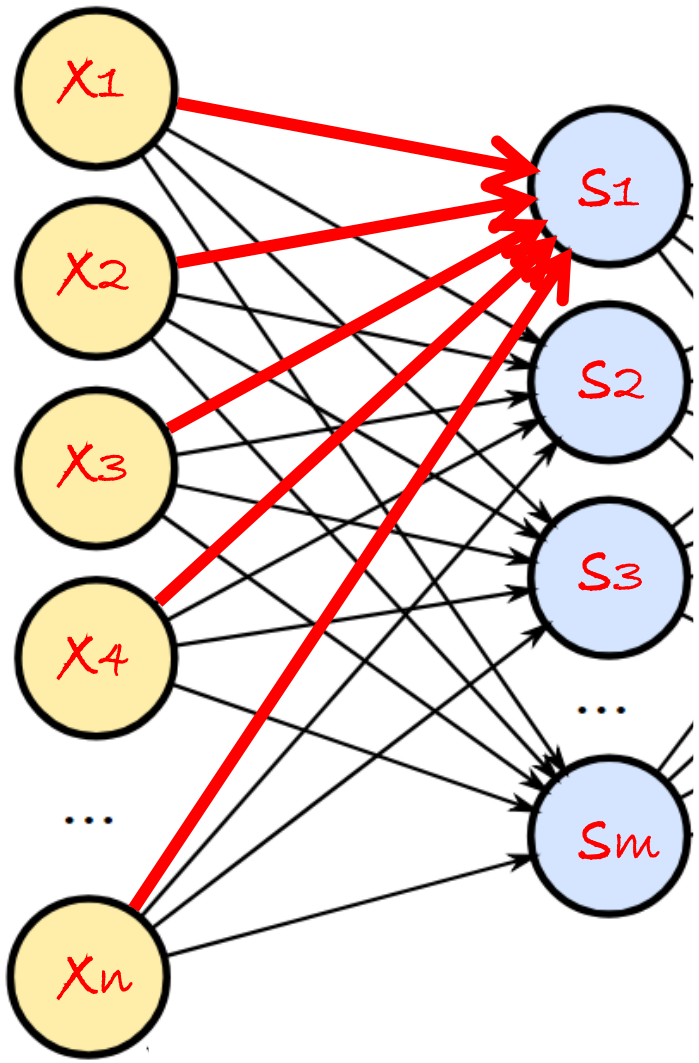

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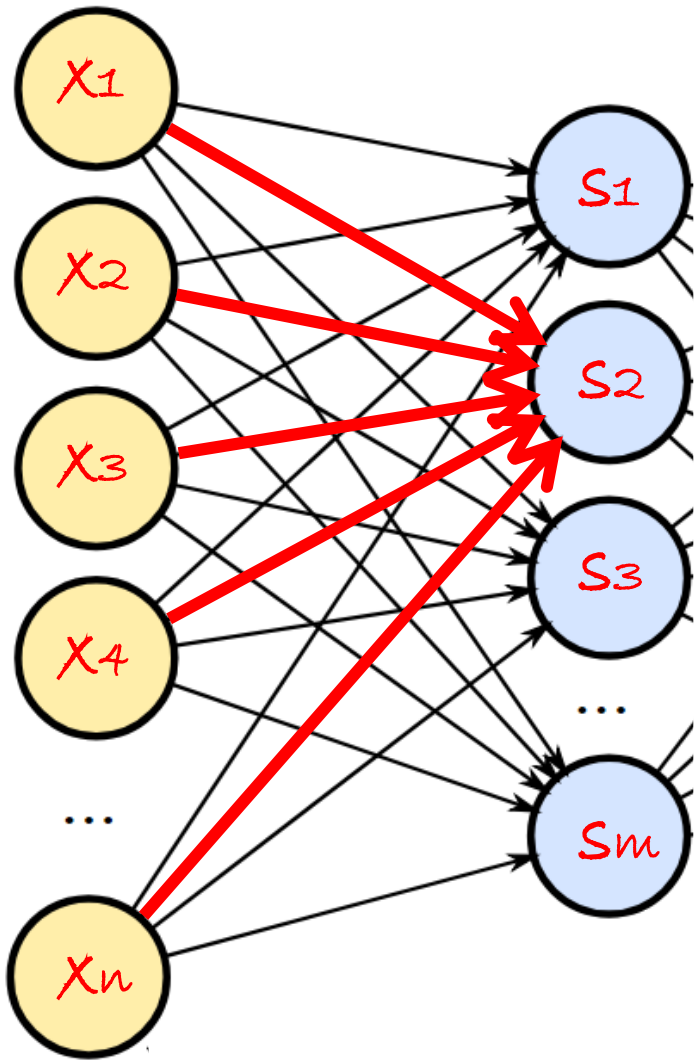

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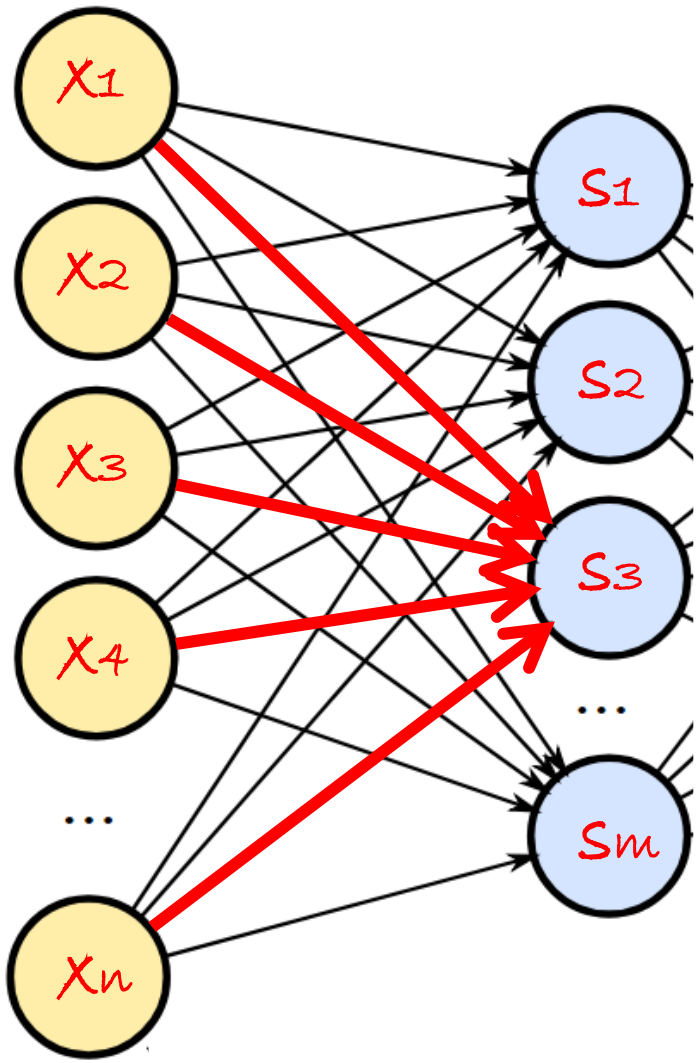

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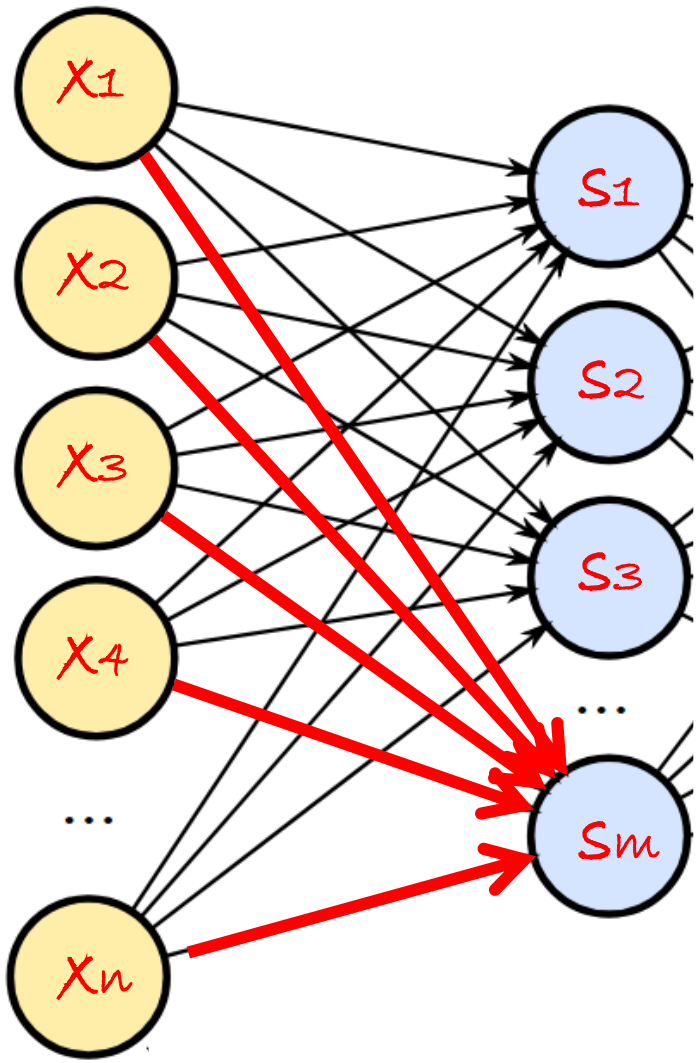

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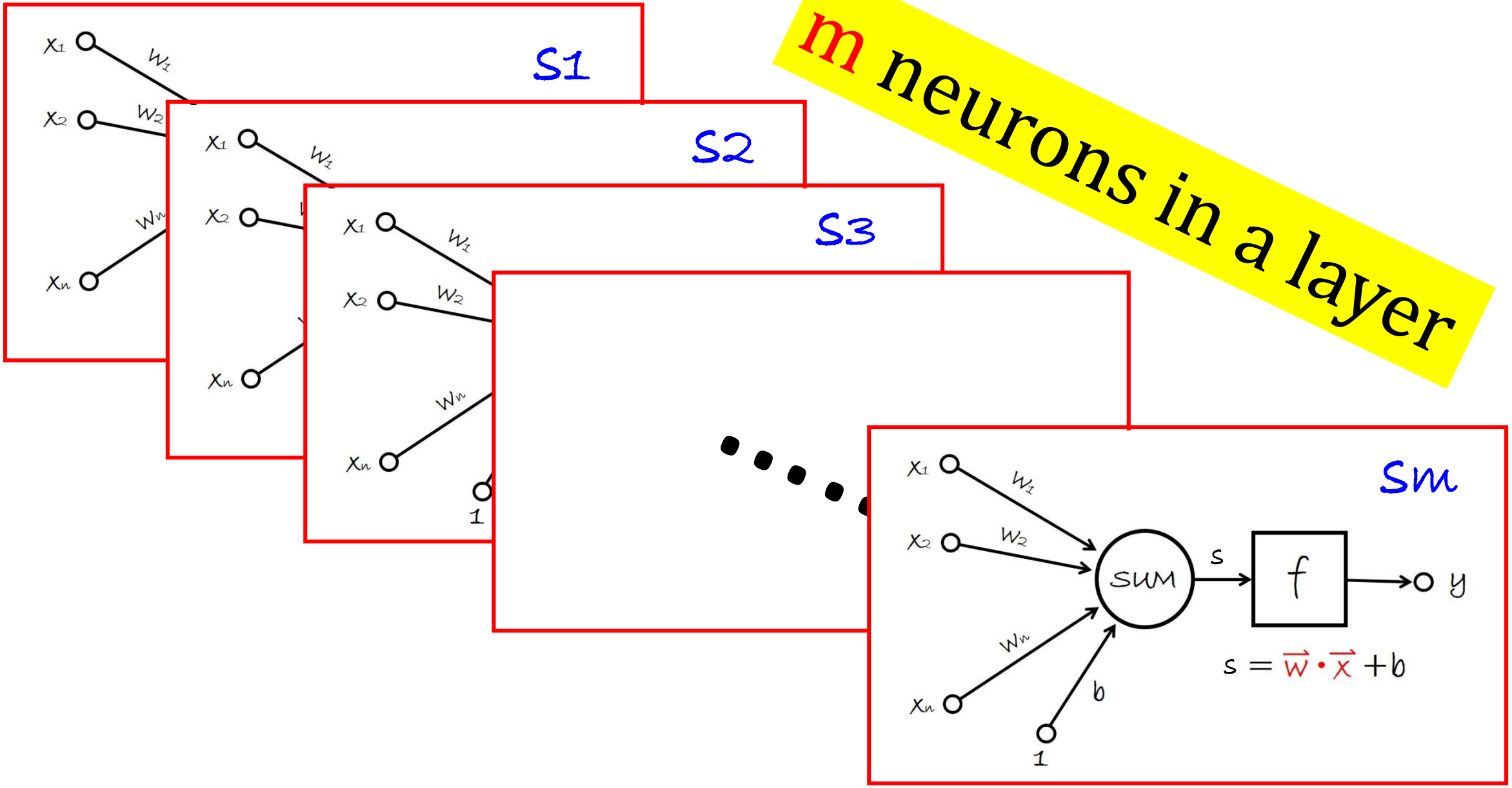
m neurons in a layer



$$w_{11}x_1 + w_{12}x_2 + \cdots + w_{1n}x_n + b_1 = s_1$$

$$w_{21}x_1 + w_{22}x_2 + \cdots + w_{2n}x_n + b_2 = s_2$$

$$\vdots \qquad\qquad \vdots$$

$$w_{m1}x_1 + w_{m2}x_2 + \cdots + w_{mn}x_n + b_m = s_m$$

n inputs: *X1, X2, ..., Xn*

$$w_{11}x_1 + w_{12}x_2 + \cdots + w_{1n}x_n + b_1 = s_1$$

$$w_{21}x_1 + w_{22}x_2 + \cdots + w_{2n}x_n + b_2 = s_2$$

$$\vdots \qquad\qquad \vdots$$

$$w_{m1}x_1 + w_{m2}x_2 + \cdots + w_{mn}x_n + b_m = s_m$$

m neurons: *s1, s2, ..., sm*

$$w_{11}x_1 + w_{12}x_2 + \cdots + w_{1n}x_n + b_1 = s_1$$

$$w_{21}x_1 + w_{22}x_2 + \cdots + w_{2n}x_n + b_2 = s_2$$

$$\vdots \qquad\qquad \vdots$$

$$w_{m1}x_1 + w_{m2}x_2 + \cdots + w_{mn}x_n + b_m = s_m$$

$$w_{11}x_1 + w_{12}x_2 + \cdots + w_{1n}x_n + b_1 = s_1$$

$$w_{21}x_1 + w_{22}x_2 + \cdots + w_{2n}x_n + b_2 = s_2$$

$$\vdots \qquad\qquad \vdots$$

$$w_{m1}x_1 + w_{m2}x_2 + \cdots + w_{mn}x_n + b_m = s_m$$

m x (n+1) parameters: $w_{11}, \ldots, w_{mn}, b_1, \ldots, b_m$

$$w_{11}x_1 + w_{12}x_2 + \cdots + w_{1n}x_n + b_1 = s_1$$

$$w_{21}x_1 + w_{22}x_2 + \cdots + w_{2n}x_n + b_2 = s_2$$

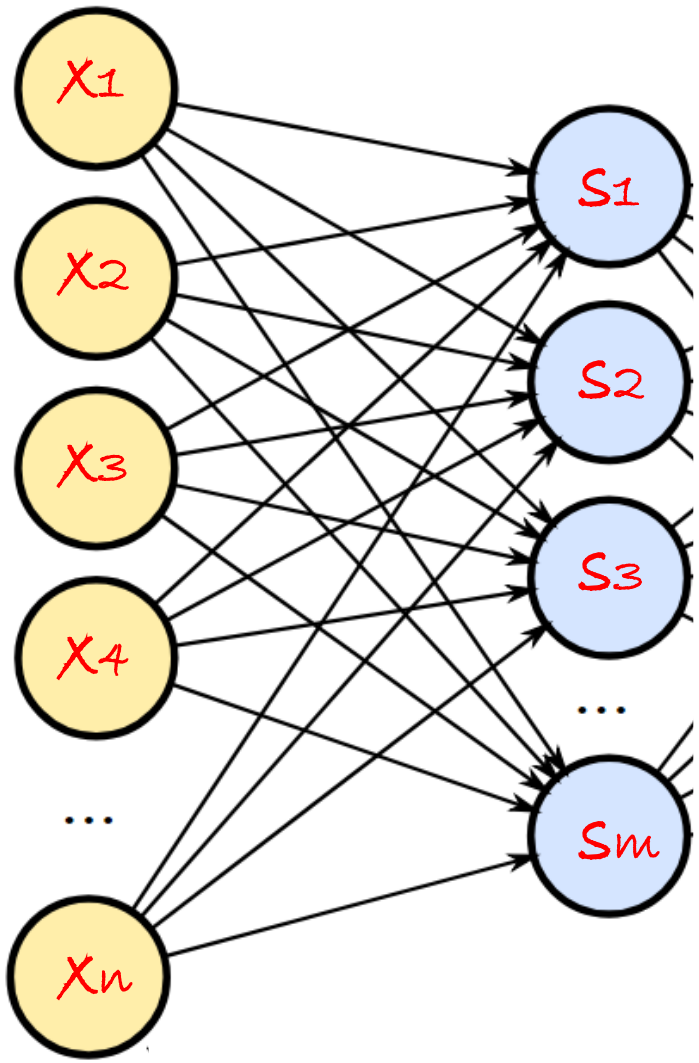
$$\vdots \qquad\qquad \vdots$$

$$w_{m1}x_1 + w_{m2}x_2 + \cdots + w_{mn}x_n + b_m = s_m$$

$$\begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & & w_{2n} \\ \vdots & & \ddots & \vdots \\ w_{m1} & w_{m2} & \cdots & w_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} s_1 \\ \vdots \\ s_m \end{bmatrix}$$

$$\underset{m\text{x}n}{W} \quad \underset{n\text{x}1}{X} + \underset{m\text{x}1}{B} = \underset{m\text{x}1}{S}$$

mx1



$$\underset{m \times n}{W}\ \underset{n \times 1}{X} + \underset{m \times 1}{B} = \underset{m \times 1}{S}$$

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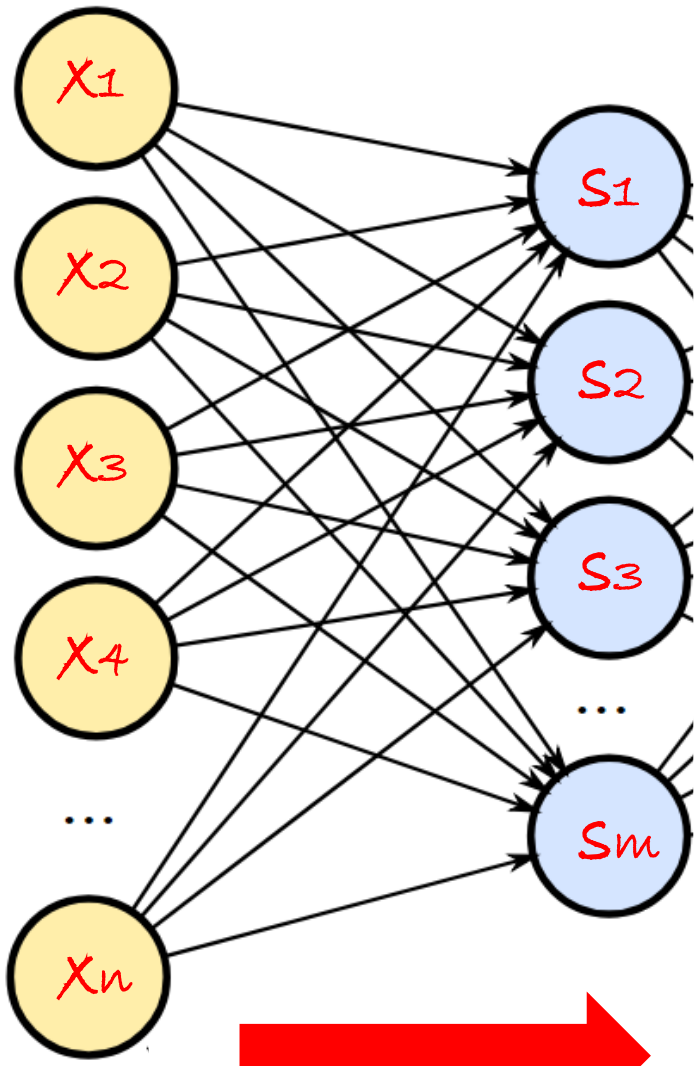

$$\underset{m \times n}{W}\ \underset{n \times 1}{X} + \underset{m \times 1}{B} = \underset{m \times 1}{S}$$

Feed Forward

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Thank you for your attention